

Codebook | Federal Transit Administration | 2023 Annual Energy Consumption and Fare Data

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Overview of Data

This data set contains information on the reported annual consumption of fuel and energy, as well as annual collected fares, for public transportation programs across the United States in the year 2023. The data is organized by agency name (i.e. the entity that manages each public transportation program, with each agency receiving a unique ID), as well as mode (form of transit service) and type of service (whether the service is directly operated by the agency itself or through a third party agency).

For each observation, the first 14 variables represent the quantity of each type of fuel or energy used in 2023 for the indicated mode of transportation reported by each transit agency. The final three variables represent fares collected for each mode of transportation in 2023, broken into fares that were paid directly by riders/users of the service and fares that were subsidized by third parties like employers, government programs, etc.

Sources and Methodology

This data set is a combination of two separate files publicly published by the Federal Transit Administration (FTA) in the National Transit Database (NTD), a public-facing repository of transit related data. Variables on energy consumption were pulled from the data set “2023 Annual Database Energy Consumption” while fare volume data was collected from from the file “2023 Annual Database Fare Revenues”.

The two original data sets were merged together based on the unique combination of agency name/ID and mode to better facilitate analysis of the interaction of fares and energy/fuel consumption on a granular basis. The data was then cleaned to standardize NA variables (or replace with 0 for numeric variables), pivot the “other fuel type” variable (which was previously populated with manually inputted fuel types) to a standard variable, and remove irrelevant or unpopulated variables from the final data set.

Variables

NTD ID

Variable Name: NTD ID

Variable Type: Numeric

Description: Unique identifier assigned to each agency by the FTA.

Agency Name

Variable Name: Agency.Name

Variable Type: Character

Description: Name of the agency that administers the transit service.

Reporter Type

Variable Name: Reporter.Type

Variable Type: Character

Description: Classification of the type of agency reporting energy/fuel usage and fare data to the FTA. Different reporter types are subject to different requirements for reporting (i.e. levels of specificity, etc.)

Reporter.Type	Count
Rural Reporter	2952
Reduced Reporter	1818
Full Reporter	1337
Separate Service	2

Reporting Module

Variable Name: Reporting.Module

Variable Type: Character

Description: Classification of the geography/urban demographic of the reporting agency (broader than reporter type).

Reporting.Module	Count
Rural	2952
Urban	2869
Tribe	288

Mode

Variable Name: Mode

Variable Type: Character

Description: The specific mode of transportation being provided.

Mode	Count
DR	3547
MB	1986
CB	249
VP	132
FB	53
CR	31
SR	27
LR	24
RB	19
HR	16

Mode	Count
MG	6
YR	6
TB	4
IP	3
TR	3
AR	1
CC	1
PB	1

Code	Transit Service
DR	Demand Response
MB	Motor Bus
CB	Commuter Bus
VP	Vanpool
FB	Ferryboat
CR	Commuter Rail
SR	Streetcar (Rail)
LR	Light Rail
RB	Rapid Bus or BRT (Bus Rapid Transit)
HR	Heavy Rail
MG	Monorail / Automated Guideway
YR	Hybrid Rail
TB	Trolleybus
IP	Inclined Plane
TR	Tram
AR	Aerial Tramway
CC	Cable Car
PB	Públic (Puerto Rico)

TOS

Variable Name: TOS

Variable Type: Character

Description: Whether a mode of transportation is agency-run or provided through a third party contractor.

TOS	Count
DO	4432
PT	1545
TX	112
TN	20

Code	Type of Service	Description
DO	Directly Operated	Agency-run service
PT	Purchased Transportation	Contractor-provided
TX	Taxi - Dedicated	Standard taxi service
TN	Taxi - Non-Dedicated	Rideshare (Uber, etc.)

Bio-Diesel

Variable Name: Bio.Diesel

Variable Type: Numeric

Description: Gallons of bio-diesel fuel used in 2023.

Min	Mean	Median	Max
0	5606.9	0	7961729

Bunker Fuel

Variable Name: Bunker.Fuel

Variable Type: Numeric

Description: Gallons of bunker fuel used in 2023.

Min	Mean	Median	Max
0	0	0	0

C Natural Gas

Variable Name: C.Natural.Gas

Variable Type: Numeric

Description: Cubic feet of natural gas used in 2023.

Min	Mean	Median	Max
0	29648.7	0	26223612

Diesel Fuel

Variable Name: Diesel.Fuel

Variable Type: Numeric

Description: Gallons of diesel fuel used in 2023.

Min	Mean	Median	Max
0	70299.5	0	24981101

Electric Battery

Variable Name: Electric.Battery

Variable Type: Numeric

Description: Kilowatt hours (kWh) of electric battery used in 2023.

Min	Mean	Median	Max
0	11686.1	0	6883170

Electric Propulsion

Variable Name: Gasoline

Variable Type: Numeric

Description: Kilowatt hours (kWh) of electric propulsion used in 2023.

Min	Mean	Median	Max
0	992138.6	0	1546269600

Ethanol

Variable Name: Ethanol

Variable Type: Numeric

Description: Gallons of ethanol used in 2023.

Min	Mean	Median	Max
0	0	0	0

Methonal

Variable Name: Methonal

Variable Type: Numeric

Description: Gallons of methanol used in 2023.

Min	Mean	Median	Max
0	0	0	0

Gasoline

Variable Name: Gasoline

Variable Type: Numeric

Description: Gallons of gasoline used in 2023.

Min	Mean	Median	Max
0	14766.5	0	3823315

Hydrogen

Variable Name: Hydrogen

Variable Type: Numeric

Description: Kilograms of hydrogen used in 2023.

Min	Mean	Median	Max
0	53.7	0	120217

Kerosene

Variable Name: Kerosene

Variable Type: Numeric

Description: Gallons of Kerosene used in 2023.

Min	Mean	Median	Max
0	0	0	0

Liquified Nat Gas

Variable Name: Liquified.Nat.Gas

Variable Type: Numeric

Description: Gallons of liquified natural gas used in 2023.

Min	Mean	Median	Max
0	58.1	0	240948

Liquified Petroleum Gas

Variable Name: Liquified.Petroleum.Gas

Variable Type: Numeric

Description: Gallons of liquified petroleum gas used in 2023.

Min	Mean	Median	Max
0	1402.8	0	1393449

Renewable Diesel

Variable Name: Renewable.Diesel

Variable Type: Numeric

Description: Gallons of renewable diesel used in 2023.

Min	Mean	Median	Max
0	802.5	0	4816970

Passenger Paid Fares

Variable Name: Passenger.Paid.Fares

Variable Type: Numeric

Description: Fares paid directly by riders in 2023 (USD).

Min	Mean	Median	Max
0	1593448	0	2703907746

Organization Paid Fares

Variable Name: Organization.Paid.Fares

Variable Type: Numeric

Description: Fares paid by third party organizations (i.e. employers, government programs, etc.) in 2023 (USD).

Min	Mean	Median	Max
0	118288.3	0	31584416

Total Fares

Variable Name: Total.Fares

Variable Type: Numeric

Description: Sum of all fares paid in 2023 (USD).

Min	Mean	Median	Max
0	1711736	909	2703907746